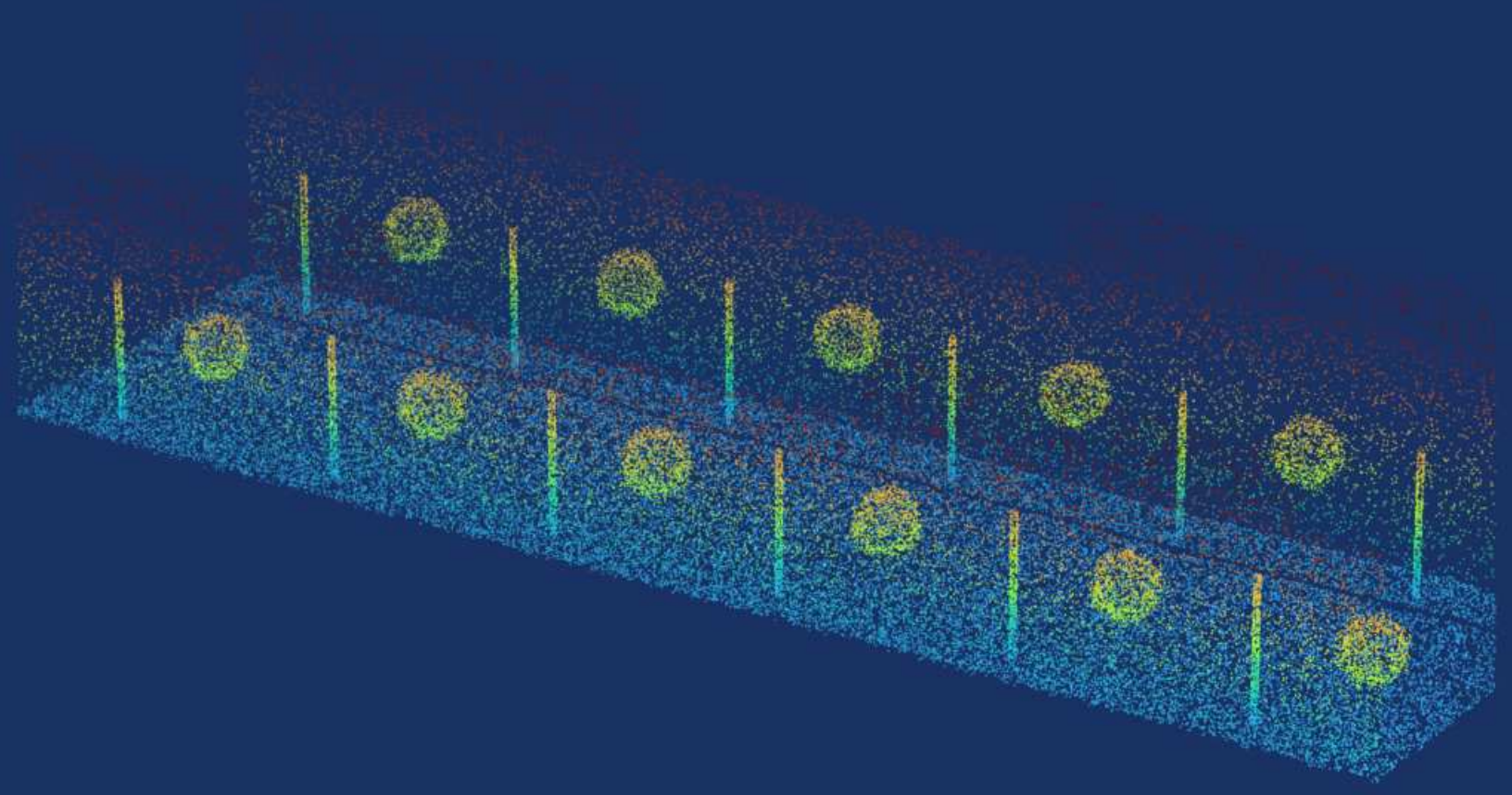
LiDAR from the Ground Up

Projects: Capstone and Hands-On Labs

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Module 21 and Appendix D of the Reference Book,
with the sample project dataset.



Companion to the Reference Book (R0)

MODULE 21

Capstone and Sample Project Dataset

This module walks through a complete mobile-LiDAR project, using a tool from every earlier module. Work through it with a calculator.

SIMPLE APPROACH

This is the "big game" at the end of the season. You use every skill you practiced: measuring with light, turning angles into dots, keeping track of the car, fixing mistakes with landmarks, and checking your work with secret checkpoints. If every step is done carefully, the final map can be trusted to build a real road.

21.1 The assignment

Project	topographic survey of a 2.0 km (6,562 ft) urban road for resurfacing design
Specification	hard-surface vertical $\text{RMSE}_z \leq 3$ cm (0.10 ft); horizontal $\text{RMSE}_H \leq 5$ cm (0.16 ft) (ASPRS Ed. 2 style)
Features	curbs, edges of pavement, striping, manholes, signs; along-track point spacing ≤ 0.25 m (0.82 ft) on pavement
Complication	a 300 m (984 ft) underpass with no GNSS
System	tilted survey-grade head, 32 channels, 20 Hz, 4 channels effectively sweep the road per rotation; tactical IMU (gyro 1°/h, accel 0.5 mg); PTP time sync; dual-frequency GNSS

21.2 Step-by-step plan

WORKED EXAMPLE — 1 — drive speed from point spacing (Module 5)

Requirement

$$\Delta_{\text{along}} = \frac{v}{f_{\text{rot}} \times N_{\text{eff}}} \leq 0.25 \text{ m}$$

Solve for v

$v \leq 0.25 \times 20 \times 4 = 20$ m/s

Choose

13 m/s (≈30 mph) for traffic and margin: $\Delta_{\text{along}} = 13/80 = 0.16$ m ✓

WORKED EXAMPLE — 2 — the underpass: how bad is an unaided IMU? (Module 7)

Outage time

$300/13 = 23.1$ s

Accelerometer drift

$\frac{1}{2} \times 0.0049 \times 23.1^2 = 1.31$ m along-track

Gyro drift

$\frac{1}{2} \times 13 \times 4.85 \times 10^{-6} \times 23.1^2 = 0.017$ m lateral

Result: lateral is fine; along-track is not. LiDAR SLAM helps but an underpass is a corridor (Module 11) with weak along-track constraint. **Plan:** chevron targets at the entrance, midpoint and exit as control, plus a return pass for loop closures.

WORKED EXAMPLE — 3 — timing and calibration checks (Modules 8–9)

Timing

PTP sync to ± 0.1 ms: $13 \times 0.0001 = 1.3$ mm — negligible

Boresight

after the morning calibration loop the residual is 0.02°; at the longest useful range (30 m): $30 \times 0.000349 = 1.0$ cm

WORKED EXAMPLE — 4 — vertical error budget (Section 14.11.4 of the Reference Book)

Open sky

2.45 cm (0.080 ft) $\leq$ 3 cm ✓

Underpass, trajectory 2.5 cm

3.00 cm (0.098 ft) — at the limit ×

Underpass with mid-point control

2.23 cm (0.073 ft) ✓

Result: the extra control target in the underpass is what makes the job pass. The full budget is in Module 14.

WORKED EXAMPLE — 5 — control and checkpoint layout (Module 13)

Checkpoints

ASPRS minimum 30 over 2.0 km: about one every 67 m, alternating sides of the road and including the underpass

Control

every ~250 m plus the three underpass targets; never reused as checkpoints

Survey accuracy

RTK/total station at $\leq$ 1.5 cm vertical, i.e. at least 2× better than the 3 cm product

21.3 Collecting and processing

1. Morning: power and storage checks, PPS/PTP lock, calibration file loaded, 5-minute static IMU alignment, short loop past known features.
2. Collect northbound and southbound at 13 m/s, log traffic and GNSS outages.
3. Office: PPK trajectory; SLAM with loop closures; inspect trajectory σ (expect a bump in the underpass); georeference; de-skew; strip-to-strip check.
4. Clean noise and traffic; classify ground; adjust to control.
5. Measure the 30 checkpoints *last*, without changing anything afterwards.

WORKED EXAMPLE — 6 — the accuracy results

Measured from 30 checkpoints: vertical $\text{RMSE}_{z,\text{data}} = 1.9$ cm (mean +0.3 cm); horizontal $\text{RMSE}_x = 2.0$, $\text{RMSE}_y = 1.7$ cm. Checkpoint survey RMSE: 0.8 cm vertical, 1.0 cm horizontal.

Vertical

$\sqrt{1.9^2 + 0.8^2} = \sqrt{3.61 + 0.64} = 2.06 \text{ cm} \leq 3 \text{ cm}$ ✓

Horizontal

$\text{RMSE}_H = \sqrt{2.0^2 + 1.7^2} = 2.62$ cm; with survey error $\sqrt{2.62^2 + 1.0^2} = 2.81 \text{ cm} \leq 5 \text{ cm}$ ✓

Cloud-to-cloud

northbound vs southbound on pavement: mean 0.2 cm, spread 0.9 cm; one 2.5 cm patch at the underpass exit flagged, re-checked against an extra shot, within tolerance

Result: the data meet specification. Write it down.

SURVEYOR'S FIELD TIP

Sample accuracy statement. "This dataset was tested for vertical accuracy at 30 independent checkpoints on hard surfaces and found to have RMSE_z = 2.1 cm, including checkpoint survey uncertainty, and RMSE_H = 2.8 cm horizontally, in accordance with the ASPRS Positional Accuracy Standards, Edition 2. Horizontal datum: [NAD83(2011) State Plane Florida East, US ft]; vertical datum: [NAVD88, GEOID18]." Fill in the datums your project actually used, and comply with the content requirements of the governing standards of practice (in Florida, Ch. 5J-17 F.A.C.).

21.4 Sample project dataset: trajectory, checkpoints and accuracy report

The following dataset is **realistic but simulated**: it was generated to match the Main Street plan above (coordinates are fictitious). It is supplied with the course as two CSV files, so participants can recompute every number:

- `sample_trajectory.csv`: time, station, position, heading, trajectory standard deviations, GNSS status;
- `sample_checkpoints.csv`: 30 checkpoints with survey and LiDAR elevations and residuals (cm and ft).

BIG IDEA

Read the trajectory before reading the point cloud. The bottom plot shows exactly where accuracy is strong (open sky, 1.4 cm / 0.05 ft) and where it is weak (underpass, up to 3.6 cm / 0.12 ft) — which is where extra checkpoints belong.

Checkpoint table

ID	Station m / ft	Side	Survey elev. m / ft	dZ cm / ft	dE cm	dN cm
CP01	33 / 108	R	6.207 / 20.365	-0.9 / -0.030	-3.0	-1.6
CP02	100 / 327	L	6.421 / 21.066	+0.1 / +0.003	-0.8	-1.6
CP03	166 / 546	R	6.627 / 21.742	+0.4 / +0.013	+3.8	+2.1
CP04	233 / 765	L	6.821 / 22.377	-0.3 / -0.010	-0.9	-1.3

ID	Station m / ft	Side	Survey elev. m / ft	dZ cm / ft	dE cm	dN cm
CP05	300 / 983	R	6.999 / 22.963	+2.8 / +0.092	-0.1	-2.3
CP06	366 / 1202	L	7.161 / 23.494	-3.6 / -0.118	+1.4	-2.7
CP07	433 / 1421	R	7.307 / 23.973	-1.0 / -0.033	+0.9	-1.1
CP08	500 / 1640	L	7.439 / 24.408	+1.7 / +0.056	-0.8	+1.5
CP09	567 / 1859	R	7.562 / 24.809	-0.5 / -0.016	-1.0	+1.5
CP10	633 / 2077	L	7.679 / 25.195	-0.4 / -0.013	+3.2	+0.5
CP11	700 / 2296	R	7.797 / 25.581	-0.6 / -0.020	-2.7	-1.5
CP12	767 / 2515	L	7.921 / 25.987	+2.4 / +0.079	-3.7	+0.9
CP13	833 / 2734	R	8.055 / 26.427	+1.8 / +0.059	+2.2	-0.7
CP14	900 / 2953	L	8.203 / 26.914	-3.0 / -0.098	+2.5	+0.6
CP15	967 / 3171	R	8.368 / 27.453	-2.5 / -0.082	-2.8	+3.0
CP16	1033 / 3390	L	8.549 / 28.047	-0.8 / -0.026	+1.6	-0.7
CP17	1100 / 3609	R	8.744 / 28.689	-0.2 / -0.007	+0.4	-2.5
CP18	1167 / 3828	L	8.952 / 29.369	+2.3 / +0.075	-0.7	+2.6
CP19	1233 / 4047	R	9.166 / 30.073	-0.1 / -0.003	-0.2	-0.6
CP20	1300 / 4265	L	9.383 / 30.783	+3.2 / +0.105	+4.0	-0.6
CP21	1367 / 4484	R	9.595 / 31.481	+4.5 / +0.148	+1.4	-2.9
CP22	1433 / 4703	L	9.800 / 32.151	-1.9 / -0.062	+1.0	+0.8
CP23	1500 / 4922	R	9.991 / 32.779	+0.5 / +0.016	+0.8	+1.3
CP24	1567 / 5141	L	10.167 / 33.357	+2.6 / +0.085	-1.8	+0.0
CP25	1634 / 5359	R	10.327 / 33.880	+0.7 / +0.023	-1.1	+1.3
CP26	1700 / 5578	L	10.470 / 34.351	+1.0 / +0.033	-0.7	-1.0
CP27	1767 / 5797	R	10.601 / 34.780	-0.9 / -0.030	-2.5	+1.7
CP28	1834 / 6016	L	10.722 / 35.178	-1.1 / -0.036	+0.9	+0.9
CP29	1900 / 6235	R	10.839 / 35.562	+0.6 / +0.020	+2.3	+2.4
CP30	1967 / 6453	L	10.958 / 35.950	+2.5 / +0.082	-0.6	-2.9
Mean (bias)				+0.31 / +0.010		
RMSE				**1.90 / 0.062**	2.00	1.71

dZ = LiDAR − survey on hard surfaces (NVA). dE, dN measured on painted chevrons and curb corners. Checkpoint survey accuracy: 0.8 cm (0.026 ft) vertical, 1.0 cm (0.033 ft) horizontal.

Computations

Vertical, data fit	sum of squares 107.99 cm^2; $\text{RMSE}_z = \sqrt{107.99/30} = 1.90$ cm (0.062 ft); mean +0.31 cm
Vertical, with checkpoint error	$\sqrt{1.90^2 + 0.8^2} = 2.06$ cm (0.068 ft)
Horizontal, data fit	$\text{RMSE}_x = 2.00$, $\text{RMSE}_y = 1.71$; $\text{RMSE}_H = \sqrt{2.00^2 + 1.71^2} = 2.63$ cm (0.086 ft)
Horizontal, with checkpoint error	$\sqrt{2.634^2 + 1.0^2} = 2.82$ cm (0.092 ft), using unrounded RMSE_H
Largest residual	4.5 cm (0.148 ft) $< 3 \times$ RMSE — no blunders

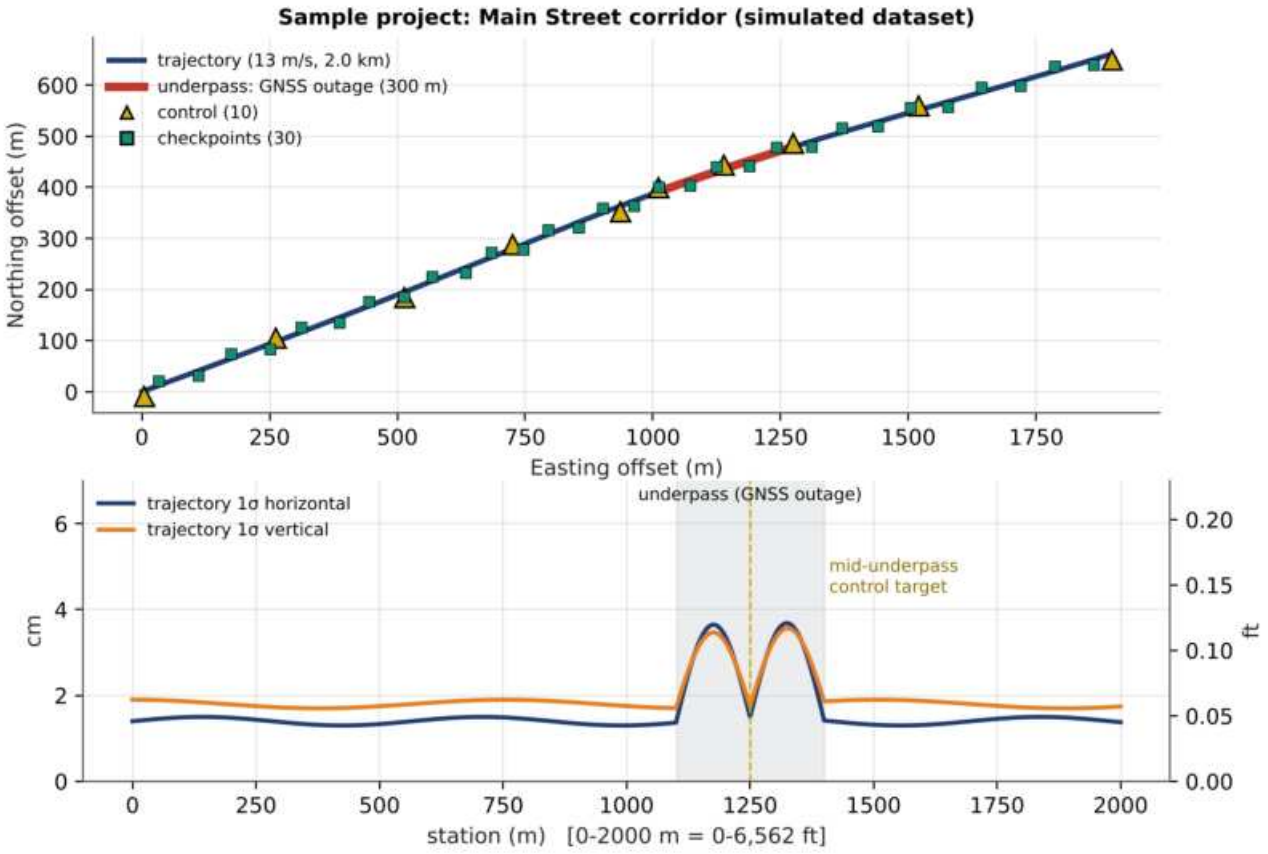


Figure 21.1: Top: the 2.0 km (6,562 ft) trajectory with 10 control points and 30 checkpoints; the red stretch is the GNSS outage under the underpass. Bottom: the trajectory's own 1σ estimates rise in the underpass and are pulled back down by the mid-underpass control target.

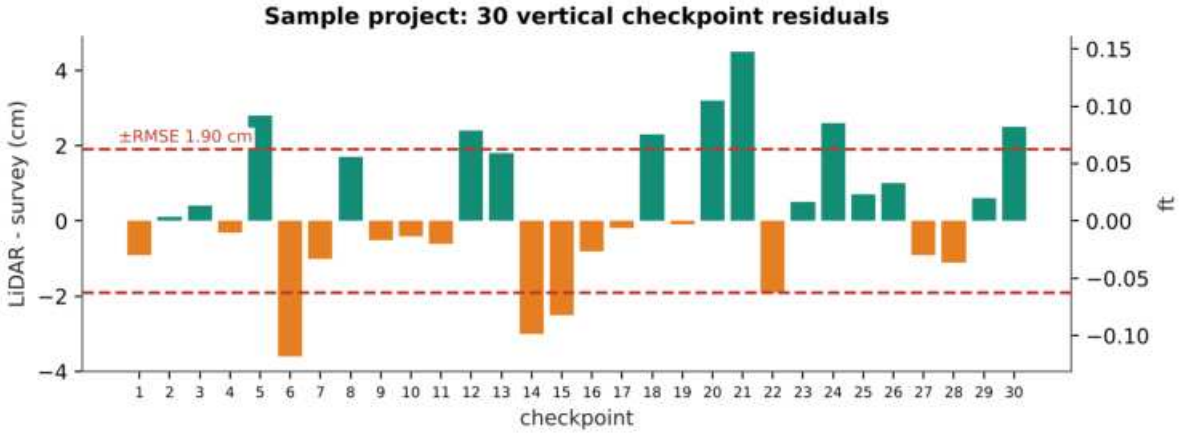


Figure 21.2: Vertical residuals at the 30 checkpoints. No residual exceeds $3\times$RMSE (5.7 cm), so none is flagged as a blunder.

Filled-in accuracy report (sample)

SURVEY AND MAP REPORT — ACCURACY SECTION (sample, simulated data)

Project	Main Street corridor topographic survey (Module 21 sample), 2.0 km / 6,562 ft
Survey type and purpose	Topographic survey by terrestrial mobile LiDAR, for roadway resurfacing design
Survey date	last date of data acquisition: _____ (5J-17.051(3)(d))
Horizontal datum	NAD 83(2011), epoch 2010.0; SPCS 83 Florida East zone; units: international foot (stated per contract)
Vertical datum	NAVD 88 via GEOID18
Control	10 control points (RTK/static GNSS and digital leveling); FPRN VRS for rover RTK; mid-underpass control target
Equipment	tilted survey-grade mobile scanner; tactical IMU; dual-frequency GNSS; PTP timing
Trajectory	PPK GNSS/INS with LiDAR SLAM and loop closures; 300 m GNSS outage bridged by SLAM and control
Accuracy agreement	3 cm $RMSE_V$ (NVA) and 5 cm $RMSE_H$ classes (Appendix E)
NVA test	30 independent hard-surface checkpoints: **$RMSE_V$ = 2.06 cm (0.068 ft)** incl. checkpoint error — **pass**
Horizontal test	30 well-defined points: **$RMSE_H$ = 2.82 cm (0.092 ft)** incl. checkpoint error — **pass**
VVA	30 vegetated checkpoints: $RMSE_V$ = 5.6 cm (0.18 ft), reported as found
Data internal precision	within-swath smooth-surface max diff 1.4 cm ($\leq$1.8); swath-to-swath $RMSD_z$ 0.9 cm ($\leq$2.4), max 3.1 cm ($\leq$4.8) — pass
Point density	median 620 points/m^2 (58 points/ft^2) on pavement within 20 m
Limitations	no data beneath vehicles parked during collection (re-shot by total station); underground utilities not mapped; points beyond 25 m excluded from design surface
Accuracy statement	"This data set was tested to meet ASPRS Positional Accuracy Standards for Digital Geospatial Data, Edition 2, Version 2 (2024) for a 3 cm $RMSE_V$ Vertical Accuracy Class and a 5 cm $RMSE_H$ Horizontal Accuracy Class. NVA was found to be $RMSE_V$ = 2.06 cm and horizontal accuracy $RMSE_H$ = 2.82 cm, including checkpoint survey uncertainty. VVA was found to be $RMSE_V$ = 5.6 cm."
Records	raw data, trajectory, calibration, control/checkpoint data and processing logs archived (5J-17.051(2)(d))
Signature and seal	PSM in responsible charge; digital deliverables sealed by signature file (5J-17.062(4))

How the sample compares with FDOT mobile LiDAR types

The report above passes the ASPRS classes agreed with the client. Resurfacing design is Type A work on an FDOT project, so the same results are converted to FDOT's 95% tolerances (Module 13, Module 19):

Item	This sample	Type A	Type B
Vertical at 95%	$1.96 \times 2.06 = 4.04$ cm (0.132 ft)	$\leq$ 0.06 ft — **fails**	$\leq$ 0.10 ft — **fails**
Horizontal at 95%	$1.7308 \times 2.82 = 4.88$ cm (0.160 ft)	$\leq$ 0.06 ft — **fails**	$\leq$ 0.10 ft — **fails**
Point density	58 points/ft^2	$\geq$ 20 — meets	$\geq$ 10 — meets
Checkpoints	30, about 220 ft apart	$\geq$ 20, $\leq$ 1000 ft — meets	$\geq$ 20, $\leq$ 2500 ft — meets
GNSS	300 m outage (about 85% of each pass fixed), bridged by SLAM and a control target	document against Note 4	document against Note 4

The survey meets its ASPRS agreement but would meet *neither* FDOT Type A nor Type B. Reaching Type A means roughly halving the RMSE, to 0.93 cm vertical and 1.06 cm horizontal: shorter ranges, slower passes, more control and checkpoints, a better IMU through the underpass, or collecting when the outage can be avoided. Settle which standard governs before mobilization (Appendix E), not after the test.

21.5 One pulse, one more time: the master checklist

#	Operation	Module
1	Range from time of flight $\rho = c\Delta t/2$ (with atmospheric correction)	3
2	Polar to sensor x, y, z	5
3	De-skew: assign each point its own pose time	8
4	Boresight rotation $\mathbf{R}_S^B$	9
5	Add lever arm $\mathbf{l}^B$	9
6	Rotate by attitude $\mathbf{R}_B^W(t)$ (roll, pitch *and* yaw)	6, 9
7	Add trajectory position $\mathbf{p}_W(t)$	9
8	Trajectory from IMU + GNSS + scan matching + loop closures in a factor graph	7, 10–12
9	Tie to control; propagate errors; test on independent checkpoints; report RMSE	13, 14
10	Clean, classify, extract, deliver, archive	18

BIG IDEA

Measure carefully, control what you can, check what you must, and document what you did. That is how survey-grade work is produced with any instrument — including a laser that never stands still.

SOURCES FOR THIS MODULE

ASPRS Edition 2, Version 2 (2024); Fla. Admin. Code R. 5J-17.051 and 5J-17.062; Appendix E (SOP and client agreement). The sample dataset is simulated for this course; its CSV files reproduce every number in the checkpoint table and report.

APPENDIX D

Hands-On Labs

These five labs turn the modules into things you can do with your hands, a tape measure, a spreadsheet or a laptop. Labs 1–2 use the simple approach and need no math; Labs 3–5 are for engineers.

Lab 1 — Be a human LiDAR (Modules 3 and 5)

SIMPLE APPROACH

You need: a protractor taped flat on a chair, a tape measure, graph paper, a friend.
Do it: Sit at the chair — you are the scanner. Point at a corner of the room, a door edge, a table leg. For each, read the angle on the protractor (that is H) and have your friend measure the distance (that is ρ). Write them in a table.
Then: On graph paper, put yourself in the middle. For each object, turn your ruler to the angle and mark a dot at the distance (1 square = 50 cm). Join the dots — you have drawn a *point-cloud map* of your room!
Engineer's extension: compute each dot with $x = \rho \cos H$, $y = \rho \sin H$ and compare with the drawing.

Lab 2 — The drifting blindfold walk (Modules 7 and 10)

SIMPLE APPROACH

Do it: Put a cone at the start. Blindfolded, try to walk 10 steps forward, turn around, and walk 10 steps back to the cone. Mark where you stop. Repeat three times. You will miss the cone — that is *drift*. Now try again, but you may touch one wall halfway (a *landmark*). Do you get closer?
What you learned: counting steps is like an IMU: little mistakes add up. Touching something you know is like a LiDAR match or a loop closure: it fixes the mistakes.

Lab 3 — ICP in 30 lines of Python (Module 11)

Run this with Python 3 and NumPy. It builds an L-shaped "room," makes a sparse, moved and turned copy (a new scan), and recovers the motion with ICP. Watch the RMS gap fall each iteration — exactly Figure 11.1.

icp_lab.py

```
import numpy as np

def best_fit(P, Q):
    """Closed-form rigid fit (Module 11): returns R, t with Q ~ R P + t."""
    pb, qb = P.mean(0), Q.mean(0)
    W = (P - pb).T @ (Q - qb)
    U, S, Vt = np.linalg.svd(W)
```

```
    R = Vt.T @ U.T
    if np.linalg.det(R) < 0:          # avoid a mirror image
        Vt[-1] *= -1; R = Vt.T @ U.T
    return R, qb - R @ pb

def icp(P, Q, iters=30):
    cur = P.copy()
    for k in range(iters):
        d = ((cur[:, None, :] - Q[None, :, :])**2).sum(-1)
        match = Q[d.argmin(1)]                # 1. nearest neighbors
        R, t = best_fit(cur, match)           # 2. best rigid fit
        cur = cur @ R.T + t                   # 3. apply
        rms = np.sqrt(d.min(1).mean())
        print(f"iter {k:2d}  RMS gap = {rms:.4f} m")
        if rms < 1e-4: break                  # 4. repeat until stable
    return best_fit(P, cur)                   # total transform

def room(n):                          # an L-shaped wall plus a short wall
    s = np.linspace(0, 1, n)
    return np.vstack([np.c_[6*s, 0*s], np.c_[6+0*s, 3*s], np.c_[3.5+0*s, 3+1.5*s]])
Q = room(600)                         # dense "map"
th = np.radians(6); Rt = np.array([[np.cos(th), -np.sin(th)], [np.sin(th), np.cos(th)]])
P = (room(40) - [0.25, -0.15]) @ Rt   # sparse new scan, moved and turned
R, t = icp(P, Q)
print("recovered angle (deg):", np.degrees(np.arctan2(R[1, 0], R[0, 0])).round(3))
```

Expected output: RMS gap 0.18 m at iteration 0 falling to about 0.002 m; recovered angle 6.0°. **Experiments:** (a) increase the turn to 25° and watch ICP get stuck in a local minimum — the reason it needs an IMU guess; (b) delete the short wall from `room()` and add a shift along x only: the long walls let the scan slide — the hallway effect.

Lab 4 — Field check of boresight from two passes (Module 9)

ENGINEER'S VIEW

Procedure: pick a sharp vertical feature beside the road — a building corner or a pole. Drive past it in both directions in the same lane. In the cloud, measure the along-track offset Δ between the feature's position in the two passes, and the range ρ from the drive line to the feature.

Why it works: a yaw boresight error $\delta\psi$ shifts the feature by $\rho\,\delta\psi$ forward on one pass and backward on the other, so the passes differ by $2\rho\,\delta\psi$:

$$\delta\psi \approx \frac{\Delta}{2\rho}.$$

Example: $\Delta = 4.2$ cm along-track at $\rho = 30$ m $\Rightarrow \delta\psi = 0.042/60 = 7.0 \times 10^{-4}$ rad $= 0.040°$. The same logic applied to the *height* of a flat road edge on both sides gives the roll residual, and to the height of features ahead/behind gives pitch.

Lab 5 — Checkpoint report in a spreadsheet (Module 13)

ENGINEER'S VIEW

Set up columns: Point ID | Survey Z | LiDAR Z | dZ = LiDAR − Survey | dZ^2. Then:

- mean: `=AVERAGE(D2:D31)`
- standard deviation: `=STDEV.S(D2:D31)`
- RMSE: `=SQRT(SUMSQ(D2:D31)/COUNT(D2:D31))`
- final RMSE with checkpoint survey error in G1: `=SQRT(RMSE^2+G1^2)`

Enter the ten residuals from the Module 13 worked example and confirm 0.53 cm, 2.64 cm, 2.56 cm and 2.75 cm. Then add a deliberate blunder (+25 cm) to one point and see how strongly RMSE reacts — and why each large residual must be investigated, not just averaged.